dwarf will gradually cool, over tens of billions of years, to become a dense, dark cinder.

Stars much heavier than the Sun burn their fuel much more quickly, and then collapse to form a neutron star or a black hole.

Neutron stars have masses about one to three times that of the Sun, circumferences of 75 to 100 kilometers (about the size of Chicago), and densities the same as the nucleus of an atom: a hundred trillion times more dense than rock and the Earth. Indeed, neutron stars are made of almost pure nuclear matter: atomic nuclei packed side by side.

Black holes (Chapter 5), by contrast, are made fully and solely from warped space and warped time (I'll explain this weird claim in Chapter 4). They contain no matter whatsoever, but they have surfaces, called "event horizons," or just "horizons," through which nothing can escape, not even light. That's why they are black. A black hole's circumference is proportional to its mass: the heavier it is, the bigger it is.

A black hole with about the same mass as a typical neutron star or white dwarf (say 1.2 times as heavy as the Sun) has a circumference of about 22 kilometers: a fourth that of the neutron star and a thousandth that of the white dwarf. See Figure 2.5.